curr_State	inputs	next State	main actions
S5oo11	data_saved=1; test_counter=2	S41000	Repeat from S41000 to S5oo11
S5oo11	test_counter =3; data_saved=1	S41000	Repeat from S41000 to S5oo11
S5oo11	test_counter =4; data_saved=1	S41000	Repeat from S41000 to S5oo11
S5oo11	test_counter =5; data_saved=1	~~Critical~~ Normal Operation	will have a ~~similar~~ critical shutdown ~~during~~ to protect circuit - Only during critical condition

* disable test & save signals

Normal operation S6 101	User_input or crit_input	crit_ shut down or norm shutdown	The shutdown will be similar however during critical shutdown it will be instantaneous goal is to protect component

* If time, will create ~~truth~~ proces of normal & critical shutdown.

ENGR202 Truth Table for the Initialization Fault report of ADAHRS LRU

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Note: Need a 2-bit counter

S0 : MCU_Init 000

S1 : Sens_Stab 001

S2 : Sens_reset 010

S3 : Conf_done 100

S4 : test_done 011

S6 : test_counter; data_saved 110

Value: $Z_2\ Z_1\ Z_0$

I = mcu_init

P = Sens_Stab

R = Sens_reset

C = Config_done

F = test_done

T1 = test_counter & T2 = test_counter

Q = data_saved, Value: $D_2\ D_1\ D_0$ are next ~~state~~ digits

State	Z_2	Z_1	Z_0	input name	input Q	C_1	C_0	D_2	D_1	D_0	next C_1	next C_0	Done
S0	0	0	0	I	0	0	0	0	0	0	0	0	0
S0	0	0	0	I									